

Experiment 8

Aim: Incorporate push notifications into your mobile application, enabling the delivery of real-time alerts or messages to users.

Theory:

Push notifications are real-time alerts or messages sent by an application to inform users about important updates and activities. In Flutter, notifications can be implemented using widgets such as `SnackBar`, `Dialog`, and `Overlay`.

In this experiment, a custom notification is displayed when the user logs into the application successfully. The app validates the username and password fields, and if the credentials are correct, a notification appears at the top of the screen using `OverlayEntry`. The notification uses animation and blur effects to create an attractive user interface.

The experiment demonstrates the use of Flutter widgets like `Overlay`, `AnimatedContainer`, `BackdropFilter`, and `SnackBar` to provide real-time user alerts and improve user interaction within the application.

Source Code:

main.dart:

```
import 'package:flutter/material.dart';
import 'dart:ui';

void main() {
  runApp(MyApp());
}

class MyApp extends StatelessWidget {
  @override
  Widget build(BuildContext context) {
    return MaterialApp(
      debugShowCheckedModeBanner: false,
      home: LoginScreen(),
    );
  }
}

class LoginScreen extends StatefulWidget {
  @override
  _LoginScreenState createState() => _LoginScreenState();
}

class _LoginScreenState extends State<LoginScreen> {
  final usernameController = TextEditingController();
  final passwordController = TextEditingController();

  void login() {
    String username = usernameController.text.trim();
    String password = passwordController.text.trim();

    if (username.isEmpty || password.isEmpty) {
      showSnack("Please fill all fields");
      return;
    }
  }
}
```

```
    if (password.length < 8) {
        showSnack("Password must be at least 8 characters");
        return;
    }

    showTopNotification(username);

    Future.delayed(Duration(seconds: 2), () {
        Navigator.push(
            context,
            MaterialPageRoute(
                builder: (_) => HomeScreen(username: username),
            ),
        );
    });
}

void showSnack(String msg) {
    ScaffoldMessenger.of(context).showSnackBar(
        SnackBar(content: Text(msg)),
    );
}

void showTopNotification(String username) {
    final overlay = Overlay.of(context);

    OverlayEntry overlayEntry = OverlayEntry(
        builder: (context) => Positioned(
            top: 60,
            left: 16,
            right: 16,
            child: Material(
                color: Colors.transparent,
                child: ClipRRect(
                    borderRadius: BorderRadius.circular(20),
                    child: BackdropFilter(
                        filter: ImageFilter.blur(sigmaX: 10, sigmaY:
10),
```

```
child: AnimatedContainer(
  duration: Duration(milliseconds: 400),
  padding: EdgeInsets.all(16),
  decoration: BoxDecoration(
    color: Colors.white.withOpacity(0.2),
    borderRadius: BorderRadius.circular(20),
    border: Border.all(color: Colors.white30),
  ),
  child: Row(
    children: [
      Icon(Icons.notifications, color:
Colors.white),
      SizedBox(width: 12),
      Expanded(
        child: Column(
          crossAxisAlignment:
CrossAxisAlignment.start,
          children: [
            Text(
              "Login Successful",
              style: TextStyle(
                fontWeight: FontWeight.bold,
                color: Colors.white),
            ),
            Text(
              "Welcome back, $username 🙌",
              style: TextStyle(color:
Colors.white70),
            ),
          ],
        ),
      ),
    ],
  ),
),
```

```
    ),  
    );  
  
    overlay.insert(overlayEntry);  
  
    Future.delayed(Duration(seconds: 2), () {  
        overlayEntry.remove();  
    });  
}  
  
@override  
Widget build(BuildContext context) {  
    return Scaffold(  
        body: Container(  
            decoration: BoxDecoration(  
                gradient: LinearGradient(  
                    colors: [Color(0xFF667eea), Color(0xFF764ba2)],  
                    begin: Alignment.topCenter,  
                    end: Alignment.bottomCenter,  
                ),  
            ),  
        child: Padding(  
            padding: EdgeInsets.symmetric(horizontal: 24),  
            child: Column(  
                mainAxisAlignment: MainAxisAlignment.center,  
                children: [  
                    Text(  
                        "Welcome to Push Notification Trial  
Application ",  
                        style: TextStyle(  
                            fontSize: 28,  
                            color: Colors.white,  
                            fontWeight: FontWeight.bold,  
                        ),  
                    ),  
                    SizedBox(height: 40),  
  
                    // Username Field
```

```
TextField(  
  controller: usernameController,  
  style: TextStyle(color: Colors.white),  
  decoration: InputDecoration(  
    hintText: "Username",  
    hintStyle: TextStyle(color: Colors.white70),  
    filled: true,  
    fillColor: Colors.white.withOpacity(0.2),  
    border: OutlineInputBorder(  
      borderRadius: BorderRadius.circular(12),  
      borderSide: BorderSide.none,  
    ),  
  ),  
)  
)  
  
SizedBox(height: 20),  
  
// Password Field  
TextField(  
  controller: passwordController,  
  obscureText: true,  
  style: TextStyle(color: Colors.white),  
  decoration: InputDecoration(  
    hintText: "Password (min 8 chars)",  
    hintStyle: TextStyle(color: Colors.white70),  
    filled: true,  
    fillColor: Colors.white.withOpacity(0.2),  
    border: OutlineInputBorder(  
      borderRadius: BorderRadius.circular(12),  
      borderSide: BorderSide.none,  
    ),  
  ),  
)  
)  
  
SizedBox(height: 30),  
  
SizedBox(  
  width: double.infinity,
```

```
        height: 50,
        child: ElevatedButton(
          onPressed: login,
          style: ElevatedButton.styleFrom(
            backgroundColor: Colors.white,
            shape: RoundedRectangleBorder(
              borderRadius: BorderRadius.circular(12),
            ),
          ),
          child: Text(
            "Sign In",
            style: TextStyle(
              color: Colors.black,
              fontWeight: FontWeight.bold,
            ),
          ),
        ),
      ),
    ),
  ],
),
);
}
```

```
class HomeScreen extends StatelessWidget {
  final String username;

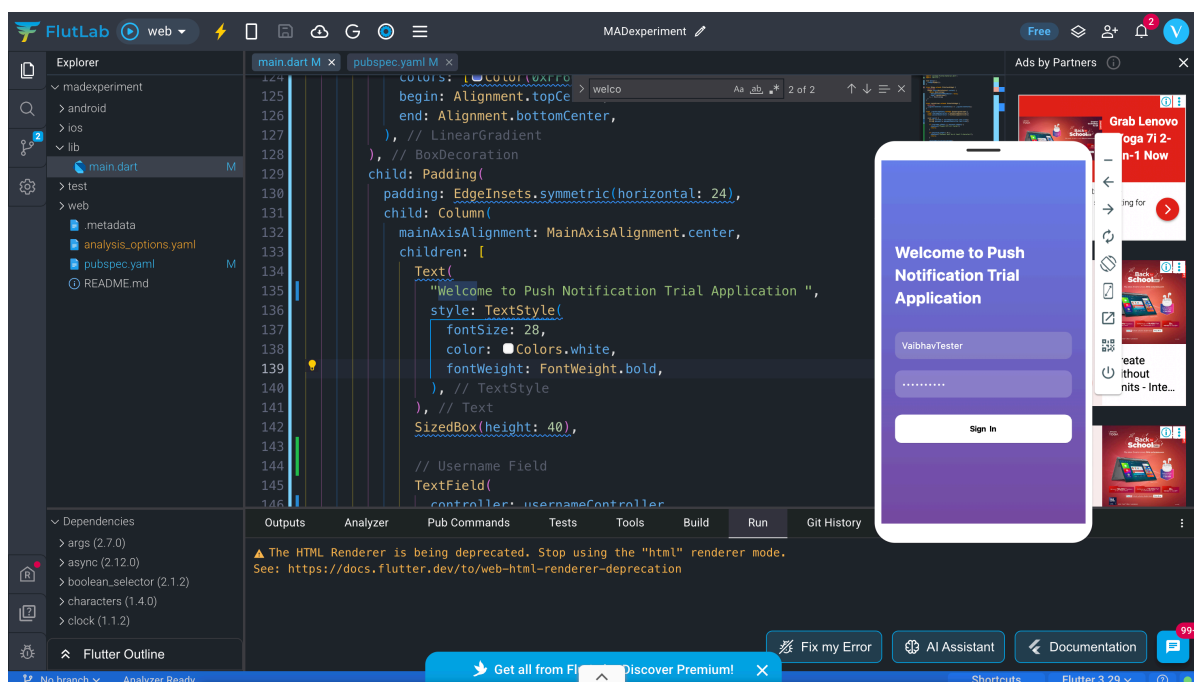
  HomeScreen({required this.username});

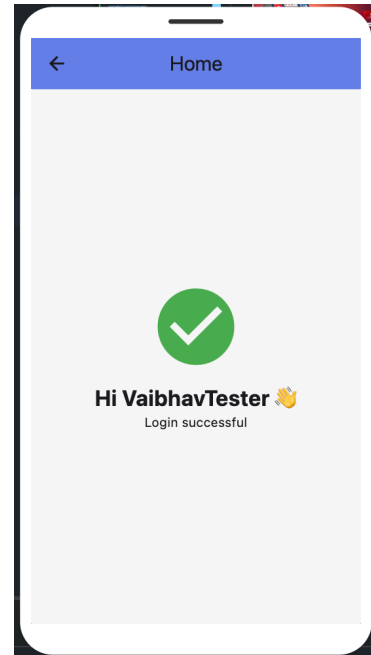
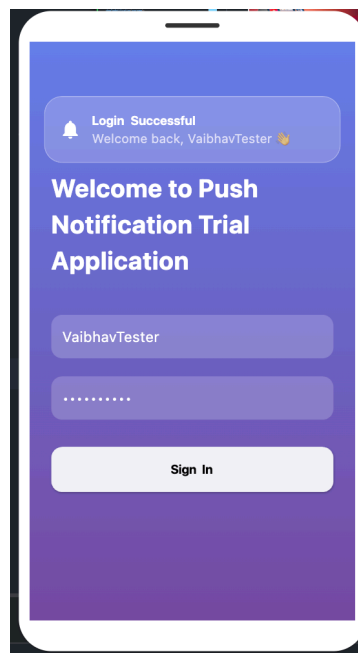
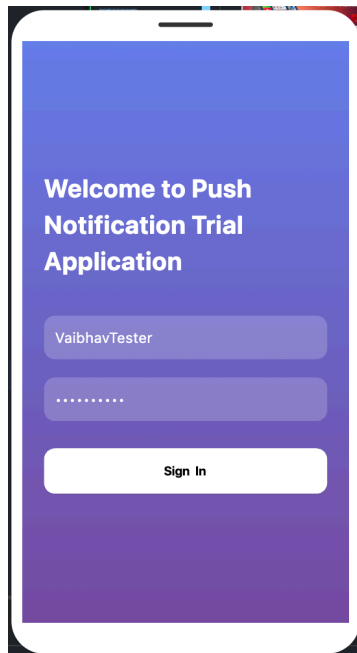
  @override
  Widget build(BuildContext context) {
    return Scaffold(
      backgroundColor: Color(0xFFf5f5f5),
      appBar: AppBar(
        title: Text("Home"),
        backgroundColor: Color(0xFF667eea),

```

```
),  
  body: Center(  
    child: Column(  
      mainAxisAlignment: MainAxisAlignment.center,  
      children: [  
        Icon(Icons.check_circle, size: 100, color:  
Colors.green),  
        SizedBox(height: 10),  
        Text(  
          "Hi $username 🙌",  
          style: TextStyle(  
            fontSize: 24,  
            fontWeight: FontWeight.bold,  
          ),  
        ),  
        Text("Login successful"),  
      ],  
    ),  
  ),  
);  
}
```

Output:





Learning Outcomes: